# Look Where It Matters: Counterfactual Visual Interventions for High-Precision Vision-Language-Action Manipulation

***Abstract*— Vision-Language-Action (VLA) models have demonstrated broad manipulation capabilities, particularly on tasks such as pick-and-place. However, they remain unreliable in high-precision manipulation, where success depends on subtle local geometry and accurate contact. A key limitation is dispersed visual attention, which often spreads across task-relevant regions, background clutter, and distractors instead of focusing on evidence critical to action generation. In contrast, humans concentrate their gaze on task-critical regions when performing precise operations. Inspired by this behavior, we introduce a training framework using counterfactual visual interventions to teach VLA policies where to look. Our method locally intervenes on visual observations and measures the resulting change in action prediction quality, deriving spatial importance directly from the policy's action objective. These action-grounded importance signals supervise a gaze module that guides the policy's visual attention toward task-relevant regions. Since the supervision is extracted from existing action demonstrations, our approach requires neither human-annotated masks or bounding boxes nor pseudo-labels from external foundation models. Extensive experiments across high-precision insertion tasks show that our method produces more concentrated, task-relevant action attention and consistently improves success rates over standard VLA. Our project page is:**

## I. Introduction

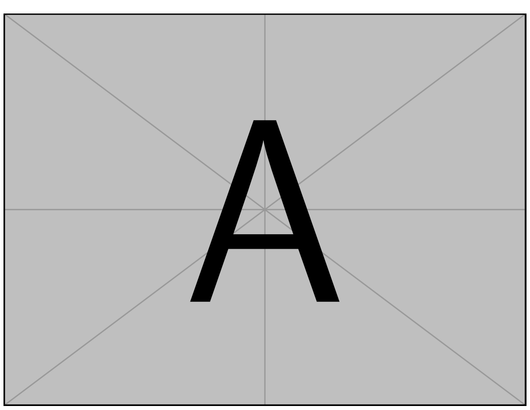


Fig. 1: Teaser: An image illustrate the main idea and some example before and after the CF intervention.

## II. Related Work

### A. Vision-Language-Action Model for Manipulation

### B. Attention Guidance for Robot Policies

### C. Causal in Visuomotor Learning

## III. Methodology

## IV. Experiment

Our experiments are designed to answer the following research questions:

**Q1.** Can our gaze model **focus more on task relevant region** and align action attention with them?

**Q2.** Does our method improve high-precision manipulation comparing to baseline, and which **key component** contribute to its performance?

**Q3.** How does our method compare with existing **visual grounding approaches**?

**Q4.** Does counterfactual gaze improve robustness of baseline to **visual distractors**?

**Q5.** Can iterative counterfactual gaze refinement **further improve the policy beyond a single training round**?

### A. Task-Relevant Region Alignment

We evaluate whether our method can discover task-relevant regions and subsequently align the visual attention used for action generation with these regions. For each task, we select 200 frames and manually annotate task-relevant regions with bounding boxes. These annotations are used only for evaluation and are never used during training.

We evaluate two spatial maps. *Gaze* is the probability map predicted by our counterfactual gaze head, representing regions whose visual intervention has a strong influence on action prediction. *Action Attention* is the attention from action tokens to image tokens in the action expert, averaged across transformer layers, attention heads, action tokens, and denoising steps. It reflects the visual evidence actually used during action generation.

Because manually annotated box boundaries can be ambiguous and the visual representation has a limited spatial resolution, we convert each annotated box $B$ into a soft ground-truth map:

$$G_\sigma(p) = \begin{cases} 1, & p \in B, \\ \exp\left(-\dfrac{d(p,B)^2}{2\sigma^2}\right), & p \notin B, \end{cases} \tag{1}$$

where $d(p, B)$ denotes the token-grid distance from $p$ to the annotated region and $\sigma = 1$ token. This soft target reduces

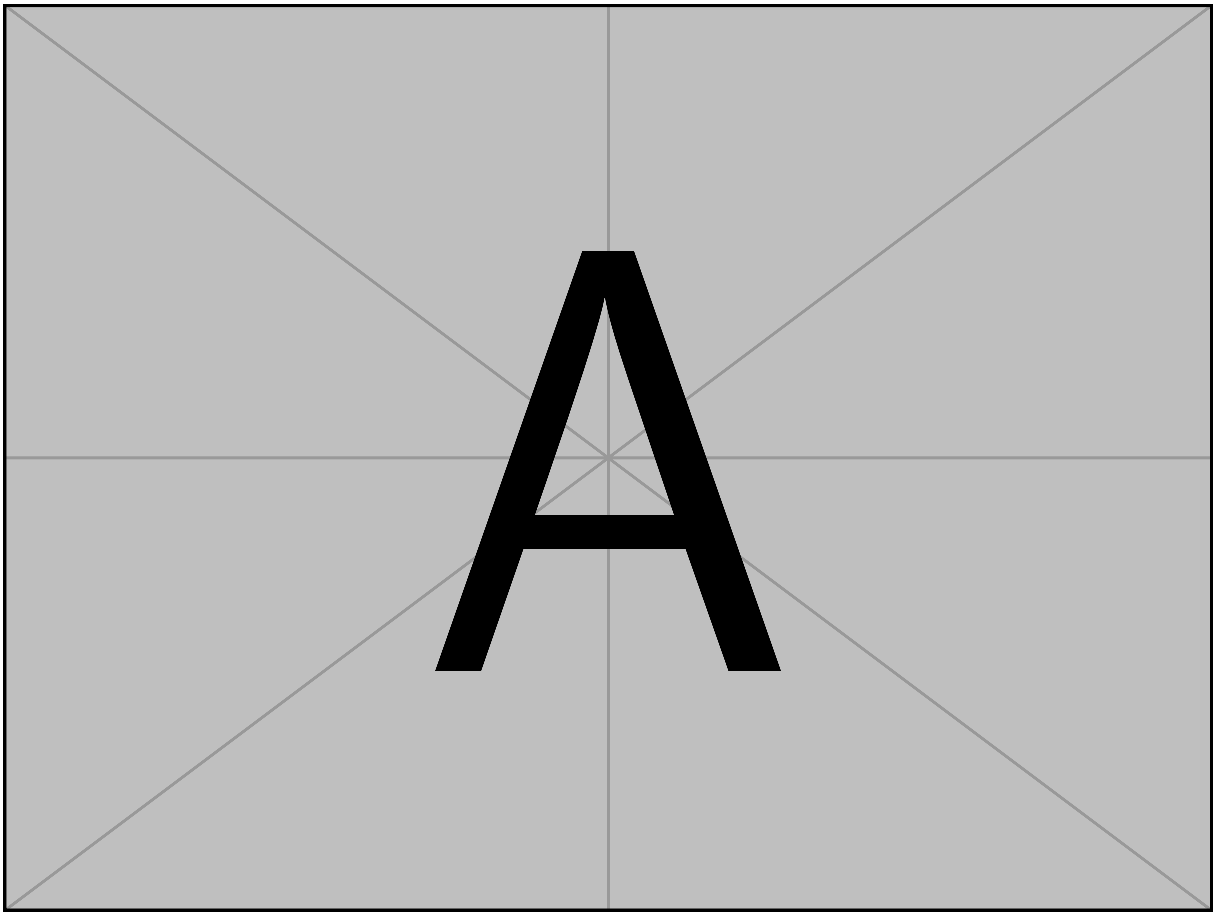

Fig. 2: Main Figure. The whole pipeline

TABLE I: Task-relevant region alignment measured by AU-Soft-AV-Prec (%).

| Method | Cable | Hose | Straw | LEGO | Avg. |
|---|---|---|---|---|---|
| $\pi_{0.5}$ Action Attn. | 17.9 | 12.0 | 14.0 | 21.2 | 16.3 |
| Ours (Gaze) | 38.7 | 33.4 | 42.5 | 46.7 | 40.3 |
| Ours (Action Attn.) | 36.9 | 30.7 | 37.3 | 38.0 | 35.7 |

evaluation bias caused by small differences in manually annotated boundaries.

Before evaluation, we normalize each attention map within its annotated camera view, remove responses below the uniform attention level $1/256$, and renormalize the remaining attention. The alignment score is then computed as

$$S = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \sum_{p} \widetilde{A}_v(p) G_v(p), \tag{2}$$

where $\widetilde{A}_v$ is the filtered attention map and $\mathcal{V}$ contains the annotated camera views. We first average frames within each episode, then average episodes within each task. The final average gives equal weight to each task.

As shown in Table I, the action attention of $\pi_{0.5}$ achieves an average alignment score of 16.3. In contrast, our counter-factual gaze achieves 40.3, demonstrating that it can discover task-relevant regions without gaze annotations or external grounding models. After grounding the action expert with the learned gaze, action attention reaches 35.7, improving over $\pi_{0.5}$ by 19.4 percentage points and retaining 88.7% of the gaze alignment. The consistent improvement across all four tasks shows that the discovered regions are effectively transferred to the visual attention used for action generation.

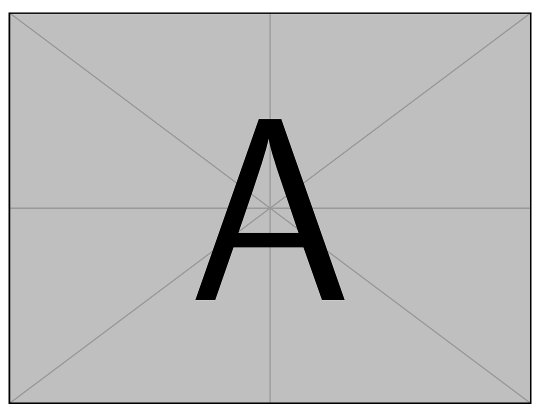

Fig. 3: Heatmap Comparison. A Comparison between base-line, ours and GroundingDINO.

## V. CONCLUSION AND LIMITATION

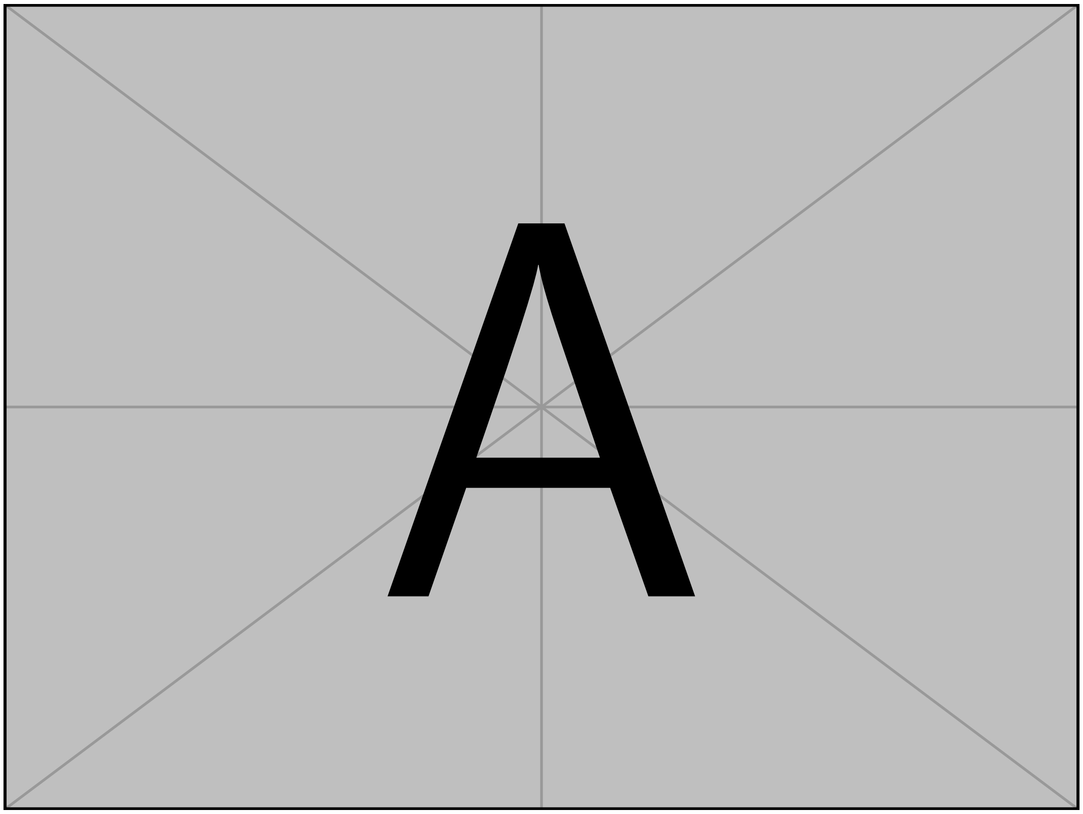


Fig. 4: Task Progress Configurations for All Four Tasks.

| Method | Straw | Hose | LEGO | Cable | Avg. |
|---|---|---|---|---|---|
| $\pi_{0.5}$ | 37/50 | 34/50 | 44/50 | – | – |
| **Ours** | **44/50** | **46/50** | **48/50** | – | – |

TABLE II: Comparison with the Baseline.

| Task | Cable | Hose | Avg. |
|---|---|---|---|
| $\pi_{0.5}$ | – | – | – |
| Action Loss Only | – | – | – |
| Stage Training | – | – | – |
| **Ours** | – | – | – |

TABLE III: Ablation of the key components on Cable insertion. GT-Mass and Conc. are computed from the final action attention.

| Method | LEGO | Straw | Avg. |
|---|---|---|---|
| $\pi_{0.5}$ | – | – | – |
| EG | – | – | – |
| CG | – | – | – |
| IG (ReconVLA-style) | – | – | – |
| **Ours** | – | – | – |

TABLE IV: Comparison with existing visual-grounding approaches on two representative high-precision manipulation tasks. All methods use the same $\pi_{0.5}$ backbone and evaluation protocol.

| Method | Clean | Clutter | Lighting | Avg. Drop ↓ |
|---|---|---|---|---|
| $\pi_{0.5}$ | 18/20 | 13/20 | 11/20 | – |
| **Ours** | **19/20** | **15/20** | 11/20 | – |

TABLE V: Robustness to Visual Distractors in LEGO Assembly.

## REFERENCES